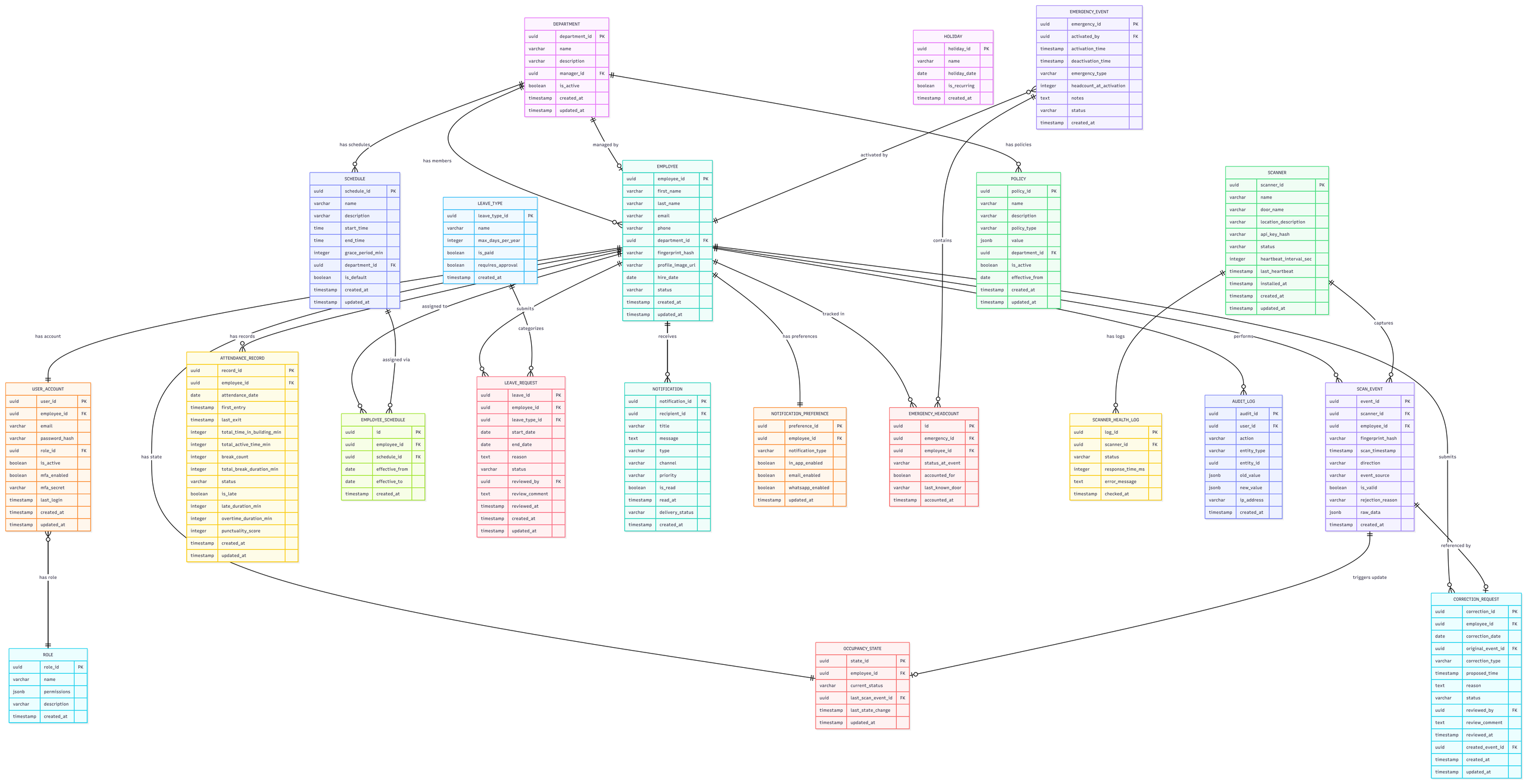
DEPARTMENT
uuid department_id PK
varchar name
varchar description
uuid manager_id FK
boolean is_active
timestamp created_at
timestamp updated_at
HOLIDAY
uuid holiday_id PK
varchar name
date holiday_date
boolean is_recurring
timestamp created_at
EMERGENCY_EVENT
uuid emergency_id PK
uuid activated_by FK
timestamp activation_time
timestamp deactivation_time
varchar emergency_type
integer headcount_at_activation
text notes
varchar status
timestamp created_at
has schedules
has members
managed by
activated by
has policies
contains
SCHEDULE
uuid schedule_id PK
varchar name
varchar description
time start_time
time end_time
integer grace_period_min
uuid department_id FK
boolean is_default
timestamp created_at
timestamp updated_at
LEAVE_TYPE
uuid leave_type_id PK
varchar name
integer max_days_per_year
boolean is_paid
boolean requires_approval
timestamp created_at
EMPLOYEE
uuid employee_id PK
varchar first_name
varchar last_name
varchar email
varchar phone
uuid department_id FK
varchar fingerprint_hash
varchar profile_image_url
date hire_date
varchar status
timestamp created_at
timestamp updated_at
POLICY
uuid policy_id PK
varchar name
varchar description
varchar policy_type
jsonb value
uuid department_id FK
boolean is_active
date effective_from
timestamp created_at
timestamp updated_at
SCANNER
uuid scanner_id PK
varchar name
varchar door_name
varchar location_description
varchar api_key_hash
varchar status
integer heartbeat_interval_sec
timestamp last_heartbeat
timestamp installed_at
timestamp created_at
timestamp updated_at
assigned to
submits
categorizes
tracked in
has preferences
receives
has account
has records
assigned via
has logs
performs
captures
submits
has state
referenced by
triggers update
has role
USER_ACCOUNT
uuid user_id PK
uuid employee_id FK
varchar email
varchar password_hash
uuid role_id FK
boolean is_active
boolean mfa_enabled
varchar mfa_secret
timestamp last_login
timestamp created_at
timestamp updated_at
ATTENDANCE_RECORD
uuid record_id PK
uuid employee_id FK
date attendance_date
timestamp first_entry
timestamp last_exit
integer total_time_in_building_min
integer total_active_time_min
integer break_count
integer total_break_duration_min
varchar status
boolean is_late
integer late_duration_min
integer overtime_duration_min
integer punctuality_score
timestamp created_at
timestamp updated_at
EMPLOYEE_SCHEDULE
uuid id PK
uuid employee_id FK
uuid schedule_id FK
date effective_from
date effective_to
timestamp created_at
LEAVE_REQUEST
uuid leave_id PK
uuid employee_id FK
uuid leave_type_id FK
date start_date
date end_date
text reason
varchar status
uuid reviewed_by FK
text review_comment
timestamp reviewed_at
timestamp created_at
timestamp updated_at
NOTIFICATION
uuid notification_id PK
uuid recipient_id FK
varchar title
text message
varchar type
varchar channel
varchar priority
boolean is_read
timestamp read_at
varchar delivery_status
timestamp created_at
NOTIFICATION_PREFERENCE
uuid preference_id PK
uuid employee_id FK
varchar notification_type
boolean in_app_enabled
boolean email_enabled
boolean whatsapp_enabled
timestamp updated_at
EMERGENCY_HEADCOUNT
uuid id PK
uuid emergency_id FK
uuid employee_id FK
varchar status_at_event
boolean accounted_for
varchar last_known_door
timestamp accounted_at
SCANNER_HEALTH_LOG
uuid log_id PK
uuid scanner_id FK
varchar status
integer response_time_ms
text error_message
timestamp checked_at
AUDIT_LOG
uuid audit_id PK
uuid user_id FK
varchar action
varchar entity_type
uuid entity_id
jsonb old_value
jsonb new_value
varchar ip_address
timestamp created_at
SCAN_EVENT
uuid event_id PK
uuid scanner_id FK
uuid employee_id FK
varchar fingerprint_hash
timestamp scan_timestamp
varchar direction
varchar event_source
boolean is_valid
varchar rejection_reason
jsonb raw_data
timestamp created_at
ROLE
uuid role_id PK
varchar name
jsonb permissions
varchar description
timestamp created_at
OCCUPANCY_STATE
uuid state_id PK
uuid employee_id FK
varchar current_status
uuid last_scan_event_id FK
timestamp last_state_change
timestamp updated_at
CORRECTION_REQUEST
uuid correction_id PK
uuid employee_id FK
date correction_date
uuid original_event_id FK
varchar correction_type
timestamp proposed_time
text reason
varchar status
uuid reviewed_by FK
text review_comment
timestamp reviewed_at
uuid created_event_id FK
timestamp created_at
timestamp updated_at

ER Diagram Description

Enterprise Real-Time Attendance & Occupancy Tracking System (ERAOTS)

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1. Overview

The ER diagram represents the database design of the **Enterprise Real-Time Attendance & Occupancy Tracking System (ERAOTS)**.

This system is designed to:

- Capture employee attendance using biometric scanners
- Track real-time occupancy status (inside/outside building)
- Manage work schedules, leave requests, and policies
- Handle emergency situations with live headcount tracking
- Maintain audit logs and system monitoring data

The design follows **normalized relational database principles** to ensure:

- Data integrity
- Scalability
- High performance
- Auditability

---

2. Core Entities

2.1 EMPLOYEE

The **EMPLOYEE** entity is the central component of the system.

It stores personal and organizational details of each employee, including:

- Name, email, and phone number
- Department association
- Employment status (ACTIVE, INACTIVE, TERMINATED)
- Biometric identifier (fingerprint_hash)

Each employee:

- Belongs to one department
- Generates multiple scan events
- Has one user account
- Has one occupancy state

- Can have multiple attendance records, leave requests, and notifications

---

2.2 DEPARTMENT

The **DEPARTMENT** entity represents organizational units.

Attributes include:

- Department name and description
- Manager reference (manager_id)
- Active status

A department:

- Can have many employees
- Can define schedules and policies
- Is managed by one employee (self-referencing relationship)

2.3 USER_ACCOUNT

The **USER_ACCOUNT** entity handles authentication and system access.

It includes:

- Login credentials (email and password hash)
- Role assignment
- Multi-factor authentication settings

Each user account:

- Is linked to exactly one employee
- Is assigned one role

2.4 ROLE

The **ROLE** entity defines system permissions.

Examples:

- SUPER_ADMIN
- HR_MANAGER
- EMPLOYEE

Permissions are stored in a flexible JSON format to allow dynamic access control.

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3. Event Tracking Entities

3.1 SCAN_EVENT

The **SCAN_EVENT** entity is the most critical component of the system.

It records every fingerprint scan event with:

- Timestamp
- Scanner used
- Employee involved
- Direction (IN/OUT)
- Validation status

Key characteristics:

- Immutable (never updated or deleted)
- Serves as the primary audit trail
- Source for all attendance calculations

3.2 OCCUPANCY_STATE

The **OCCUPANCY_STATE** entity maintains the current real-time status of each employee.

It includes:

- Current status (ACTIVE, ON_BREAK, OUTSIDE, etc.)
- Reference to the last scan event
- Timestamp of last update

Each employee has exactly one occupancy state record, which is updated whenever a valid scan occurs.

3.3 ATTENDANCE_RECORD

The **ATTENDANCE_RECORD** entity stores processed daily attendance summaries.

It includes:

- First entry and last exit times
- Total working hours
- Break duration

- Late and overtime calculations
- Punctuality score

These records are derived from scan events and used for reporting and analytics.

---

4. Schedule and Leave Management

4.1 SCHEDULE

Defines work shifts with:

- Start and end times
- Grace periods

Schedules can be:

- Department-specific
- Global (default schedules)

4.2 EMPLOYEE_SCHEDULE

A junction table that enables a many-to-many relationship between employees and schedules.

It allows:

- Assigning different schedules to employees
- Managing schedule changes over time using effective dates

4.3 LEAVE_TYPE

Defines categories of leave such as:

- Annual leave
- Sick leave

Includes rules like:

- Maximum days per year
- Whether approval is required

4.4 LEAVE_REQUEST

Represents employee leave applications.

Tracks:

- Leave duration
- Reason
- Approval status (PENDING, APPROVED, REJECTED)
- Reviewer details

4.5 HOLIDAY

Stores company-wide holidays.

Supports:

- One-time holidays
- Recurring annual holidays

---

5. Notification System

5.1 NOTIFICATION

Logs all notifications sent to employees.

Includes:

- Title and message
- Channel (Email, WhatsApp, In-App)
- Priority level
- Delivery status

5.2 NOTIFICATION_PREFERENCE

Stores employee-specific notification settings.

Allows users to:

- Enable/disable channels
- Customize notification types

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6. Correction and Policy Management

6.1 CORRECTION_REQUEST

Handles attendance corrections.

Used when:

- A scan is missed
- Incorrect scan occurs

Tracks:

- Original event
- Proposed correction
- Approval workflow

6.2 POLICY

Stores configurable business rules.

Examples:

- Grace period duration
- Overtime thresholds
- Break limits

Policies are stored in JSON format for flexibility and scalability.

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7. Emergency Management

7.1 EMERGENCY_EVENT

Represents an emergency situation.

Tracks:

- Activation and deactivation time
- Type of emergency
- Initial headcount

7.2 EMERGENCY_HEADCOUNT

Tracks each employee during an emergency.

Includes:

- Whether the employee is accounted for
- Last known location
- Status during the event

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8. Monitoring and Auditing

8.1 SCANNER

Represents biometric devices.

Stores:

- Device status
- Location
- API authentication details

8.2 SCANNER_HEALTH_LOG

Maintains logs of scanner performance.

Used for:

- Monitoring uptime
- Detecting failures

8.3 AUDIT_LOG

Tracks all administrative actions.

Includes:

- Action performed
- Entity affected
- Before/after values

Ensures:

- Accountability
- Compliance

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9. Relationships Summary

The system includes the following key relationships:

- **One-to-Many**
    - Department → Employees
    - Employee → Scan Events
    - Employee → Attendance Records

- **One-to-One**
  - Employee → User Account
  - Employee → Occupancy State
- **Many-to-Many**
  - Employee ↔ Schedule (via EMPLOYEE_SCHEDULE)

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10. Design Characteristics

This ER model is designed with the following principles:

- **Normalization**: Eliminates redundancy
- **Separation of Concerns**:
  - Raw data (SCAN_EVENT)
  - Processed data (ATTENDANCE_RECORD)
- **Auditability**: Immutable event logs
- **Scalability**: JSON-based flexible fields
- **Real-Time Capability**: Occupancy tracking

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11. Conclusion

The ERAOTS ER diagram provides a comprehensive and scalable design for managing employee attendance and real-time occupancy tracking.

It integrates multiple functional areas including:

- Biometric data capture
- Attendance processing
- Leave and scheduling
- Emergency management
- System monitoring and auditing

This design ensures high reliability, flexibility, and performance, making it suitable for enterprise-level deployment.